St. Joseph's College of Engineering and Technology, Palai

Department of Computer Applications

24SJMCA245 – Mini Project

QR based navigation app

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SYNOPSIS

Module:1. Introduction

Educational institutions are expanding rapidly with multiple academic blocks, laboratories, libraries, seminar halls, administrative offices, and classrooms spread across different buildings and floors. As the campus infrastructure becomes more complex, students, faculty members, parents, and visitors often face difficulties locating classrooms and departments. This problem becomes more significant during admission periods, examinations, seminars, workshops, and orientation programs when thousands of new students and visitors enter the campus without prior knowledge of the building layout.

Traditional navigation methods such as printed maps, notice boards, and signboards are often insufficient because they provide only static information and require users to interpret directions manually. These methods become ineffective in large campuses where multiple buildings have similar layouts or where classrooms are frequently relocated due to timetable changes. As a result, users waste valuable time searching for destinations, leading to confusion, stress, and delays.

The proposed **QR Code Based Classroom Navigation System** addresses these challenges by integrating QR code technology with digital indoor navigation. QR codes are strategically placed at different locations such as building entrances, corridors, staircases, elevators, and junctions. When a user scans a QR code using the mobile application, the system automatically identifies the user's current position. The user then selects the desired classroom or department, and the application calculates the shortest and most efficient path using graph-based pathfinding algorithms such as Dijkstra's Algorithm or A* Search Algorithm.

The application provides real-time visual navigation with directional guidance, interactive floor maps, estimated walking distance, and step-by-step instructions until the destination is reached. The system eliminates dependence on manual directions while improving navigation efficiency throughout the campus.

Furthermore, the application includes an administrative module through which campus administrators can update classroom locations, building layouts, QR code mappings, and navigation paths without affecting user accessibility. The system is designed to be scalable, allowing future integration with GPS-assisted outdoor navigation, attendance management, timetable synchronization, emergency evacuation guidance, and accessibility features for differently-abled users.

Module:2. Problem Statement

Students, faculty members, and visitors frequently experience difficulty locating classrooms and other facilities within large educational campuses. Existing navigation methods rely on signboards, printed maps, or verbal assistance, which are often

inaccurate, outdated, or insufficient. These methods increase travel time, reduce operational efficiency, and create inconvenience, particularly for first-time visitors and newly admitted students.

There is a need for an intelligent indoor navigation system capable of identifying the user's current location and providing optimized, real-time navigation to the selected classroom using QR code technology.

Module:3. Objectives

Primary Objective

To develop a QR Code Based Classroom Navigation System that enables users to locate classrooms accurately and efficiently through digital indoor navigation.

Specific Objectives

- Develop a secure user authentication system.
- Implement QR code scanning for location identification.
- Provide interactive digital maps of campus buildings.
- Calculate the shortest navigation path.
- Display step-by-step navigation instructions.
- Maintain updated classroom and building information.
- Provide administrative control for campus management.
- Reduce the time required to locate classrooms.
- Improve campus accessibility for visitors and new students.
- Create a scalable system for future smart campus applications.

Module:4. Scope of the Project

The proposed system is designed primarily for educational institutions including colleges, universities, technical institutes, and research centers.

The application can be utilized by:

- Students
- Faculty members
- Administrative staff
- Parents
- Visitors
- Examination candidates

- Event participants

The system supports:

- Indoor navigation
- Classroom search
- Department search
- Floor-wise navigation
- QR-based location detection
- Interactive campus maps
- Route optimization
- Administrative map management

Future versions may include:

- Voice-assisted navigation
- AI-powered route optimization
- Accessibility support
- Emergency evacuation routing
- Attendance integration
- Timetable synchronization
- Event-based navigation
- Visitor management
- Parking navigation

Module:5. Existing System

Currently, most colleges depend on conventional navigation techniques.

These include:

- Printed campus maps
- Direction signboards
- Manual guidance from staff
- Classroom notice boards
- Department inquiry counters

Limitations

- Static information
- No real-time navigation

- Time-consuming
- Confusing for new students
- Difficult to update
- No shortest path calculation
- No digital guidance
- Limited accessibility

Module:6. Proposed System

The proposed QR Code Based Classroom Navigation System introduces a smart indoor navigation solution that combines QR code technology, digital maps, and shortest-path algorithms.

Users simply scan a QR code available at their current location. The application determines their position and displays the optimal route to the selected destination.

The navigation process includes:

- Location identification
- Destination selection
- Route calculation
- Interactive map visualization
- Step-by-step guidance
- Destination notification

The administrator manages all classroom information, maps, QR codes, and navigation routes through an integrated admin panel.

Module:7

1. User Authentication Module

This module manages user registration, login, authentication, password management, and profile information while ensuring secure access to the system.

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2. QR Code Scanning Module

Responsible for scanning QR codes placed throughout the campus.

Functions include:

- QR recognition
- Current location detection

- Data decoding
- Location verification
- Navigation initialization

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3. Classroom Search Module

Allows users to search classrooms using:

- Room number
- Faculty name
- Department
- Building
- Floor

Displays classroom information and destination details.

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4. Indoor Map Module

Displays interactive digital floor maps.

Functions include:

- Building visualization
- Floor switching
- Zoom controls
- Current location display
- Destination highlighting

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5. Path Finding Module

Calculates the shortest route using graph traversal algorithms.

Algorithms:

- Dijkstra Algorithm
- A* Search Algorithm

Factors considered:

- Distance
- Connectivity
- Accessibility
- Floor transitions

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6. Navigation Guidance Module

Provides navigation instructions through:

- Direction arrows
- Walking path
- Turn indicators
- Destination alerts
- Progress tracking

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7. Database Management Module

Stores:

- User information
- QR code data
- Classroom information
- Building layouts
- Navigation paths
- Floor maps
- Administrative records

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8. Admin Module

Allows administrators to:

- Add classrooms
- Modify classroom details
- Generate QR codes
- Upload maps
- Update navigation routes
- Monitor application usage
- Maintain system database

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8. Technologies Used

Frontend:

- Flutter

- Dart
- HTML
- CSS
- JavaScript
- React Native (optional)

Backend:

- Node.js
- Express.js
- Django
- Spring Boot

Database:

- MySQL
- Firebase Firestore

QR Code Technology:

- ZXing Library
- Flutter QR Scanner
- Google ML Kit Barcode Scanner

Navigation Technologies:

- Google Maps SDK
- Indoor Mapping APIs
- Dijkstra Algorithm
- A* Algorithm

Development Environment:

- Android Studio
- Visual Studio Code
- Git
- GitHub

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9. Hardware Requirements

Client Side:

- Android Smartphone
- Camera

- Minimum 4 GB RAM
- Internet Connection

Server Side:

- Intel Core i5 Processor or above
- 8 GB RAM
- 500 GB Storage
- Cloud or Local Server

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10. Software Requirements

- Android Studio
- Flutter SDK
- Dart SDK
- Node.js
- MySQL
- Firebase
- Visual Studio Code
- GitHub
- Windows 10/11 or Linux

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11. Expected Outcomes

The proposed system is expected to:

- Reduce navigation time within the campus.
- Improve user experience through digital guidance.
- Minimize confusion during examinations and admissions.
- Provide accurate classroom location services.
- Enhance campus management through centralized administration.
- Increase operational efficiency.
- Support future smart campus initiatives.